

In a **Bundled Grading** system, requirements are grouped into "bundles" that correlate with specific letter grades. To earn a grade, a student must complete **all** specifications within that bundle and all previous (lower) bundles.

This approach rewards complexity and depth, allowing students to choose the level of challenge they wish to tackle.

Bioinformatics Proposal: Bundle Specifications

Bundle C: The Foundational Level

Minimum requirements to demonstrate basic competency.

- **Biological Problem:** Clearly identifies a biological question and provides a basic explanation of its importance.
- **Data Description:** Identifies the available database and the specific dataset to be used.
- **Coding Approach:** Lists the primary programming language and at appropriate libraries/packages/pipelines. Provides a high-level description of the analysis.
- **Outcomes:** Identifies at least one expected deliverable (e.g., a plot or table).
- **Timeline:** Includes a basic three-step timeline (Start, Middle, End).
- **Formatting:** Proposal is 3–5 pages, excluding figures and tables, legible, and includes a reference list.

Bundle B: The Practitioner Level

Includes all requirements from Bundle C, plus the following:

- **Depth of Problem:** Includes a specific, testable hypothesis or question supported by at least three peer-reviewed citations.
- **Data Rigor:** Includes a description of data "cleaning" or preprocessing steps (e.g., filtering low-quality reads or normalizing counts).
- **Coding Sophistication:** Includes a flow-chart or detailed step-by-step logic of the pipeline. Mentions specific statistical tests or algorithms to be used.
- **Interpretation:** Explains how the results will specifically address the hypothesis or question (the "so what?" factor).
- **Timeline:** Includes a detailed breakdown with specific weekly milestones.

Bundle A: The Researcher Level

Includes all requirements from Bundles C and B, plus the following:

- **Reproducibility:** Includes a section on computational reproducibility, specifying how environments will be managed (e.g., Conda, Docker, or a specific GitHub repository structure).
- **Advanced Analysis:** Proposes a secondary analysis or a "Plan B" (e.g., "If the differential expression yields no results, I will perform a Gene Set Enrichment Analysis").

- **Critical Evaluation:** Provides a robust discussion of potential pitfalls, technical limitations, and how biological noise might impact the findings.
- **Professionalism:** Writing is of "submission quality," requiring zero to minimal editing, with perfectly formatted citations and high-resolution conceptual diagrams (if any).

The "Unsatisfactory / Fail" Category

A grade of **F** (or Unsatisfactory) is assigned if the proposal meets any of the following conditions:

- **Incomplete:** Fails to meet the basic requirements of **Bundle C**.
- **Plagiarism:** Contains unoriginal work or fails to cite data sources and literature.
- **Technical Vacuity:** The "coding approach" is too vague to be implemented (e.g., "I will use Python to look at the data" without specifying libraries or methods).
- **Formatting Breach:** Exceeds the 5-page limit or is significantly under the length required to address the specs.
- **Scope:** The project is either trivial (can be done in 10 minutes) or impossibly large (requires a multi-year PhD to complete).

Grading Key

Bundle Completed	Final Grade
Bundle A	90+
Bundle B	80-90
Bundle C	70-80
Does not meet Bundle C	<70